

Pipette and Paper: Combining molecular and genealogical methods to assess a Nile tilapia (*Oreochromis niloticus*) breeding program

Maria G. Nayfa^{1,2*}, David B. Jones¹, Curtis E. Lind^{3,5}, John A.H. Benzie^{3,6}, Dean R. Jerry^{1,2,4}, and Kyall R. Zenger¹

¹ Centre for Sustainable Tropical Fisheries and Aquaculture, College of Science and Engineering, James Cook University, Townsville, QLD, 4811 Australia

² Centre for Tropical Bioinformatics and Molecular Biology, College of Science and Engineering, James Cook University, Townsville, QLD, 4811 Australia

³ WorldFish, Jalan Batu Maung, Bayan Lepas 11960, Penang, Malaysia

⁴ Tropical Futures Institute, James Cook University, Singapore

⁵ Commonwealth Scientific and Industrial Research Organisation, Castray Esplanade, Battery Point TAS 7004 Australia

⁶ School of Biological, Earth and Environmental Sciences, University College Cork, Cork, Ireland.

* Corresponding author details:

Maria G. Nayfa, Building 32, James Cook University, Townsville, 4811 QLD, Australia.

+61 7 4781 6067

mnayfa12@gmail.com

Co-author email:

David B. Jones: david.jones3@jcu.edu.au

Curtis E. Lind: curtis.lind@csiro.au

John A. H. Benzie: j.benzie@cgiar.org

Dean R. Jerry: dean.jerry@jcu.edu.au

Kyall R. Zenger: kyall.zenger@jcu.edu.au

1. Introduction

Aquaculture selective breeding programs employ a closed nucleus mating strategy whereby animals displaying sought-after characteristics are mated to produce next generation offspring with increased prevalence of desirable phenotypes. Offspring exhibiting high genetic merit for favorable traits are then usually chosen as candidate parents for the subsequent breeding cycle. The selective breeding process is replicated each succeeding generation in order to accumulate genetic gain within the breeding population. The long term success of these closed breeding systems is dependent on a number of factors: including, the heritability of a trait, the intensity of selection, the additive genetic variance observed in the founding population, and the amount of additive genetic variance maintained over subsequent generations (Falconer et al., 1996, Loughnan et al., 2016). If breeding practices are not properly managed, the number of animals with high relatedness will increase (Gjedrem and Baranski, 2009b), leading to a substantial loss of genetic diversity over subsequent generations (Pante et al., 2001). The maintenance of genetic diversity is critical to accommodate current and future changes in production environments, and if left unchecked, it can lead to inbreeding through increased homozygosity and deleterious fitness consequences associated with inbreeding depression (Pante et al., 2001). This loss of genetic diversity can also hamper progress within the selective breeding program as it limits the amount of genetic variance available for selection (Falconer, 1960).

Aquaculture selective breeding programs may be particularly vulnerable to both a loss of genetic variation and inbreeding due to founder effects, genetic drift, high selection intensity, differential survival and parental contribution (Boudry et al., 2002, Frost et al., 2006, Lind et al., 2010). This increased susceptibility to a loss of genetic variation and inbreeding is often due to the life histories of many aquatic animals: including, asynchronous spawning (Nguyen, 2015); larval

sizes below minimum sizes for physically tagging individuals (Ouedraogo et al., 2014); and high fecundity, which encourage the use of fewer founding individuals within a selective breeding program (Gjedrem and Baranski, 2009a). To counteract these effects and maintain genetic diversity while maximizing genetic gains, aquaculture selective breeding programs rely on the ability to trace pedigrees to assess relatedness to manage family lines based on founders. As the selection program progresses, family lines share a greater and greater co-ancestry with one another and the potential for inbreeding increases (Gjedrem and Baranski, 2009b). Additionally, shared co-ancestry amongst families occurs more rapidly in selective breeding programs because the selection of individuals from each family is not random, and they likely contain favorable quantitative trait loci from the same few high performing founders (Sonesson et al., 2012).

Traditionally, genealogical data has been utilized to monitor inbreeding rates, relatedness, and co-ancestry within selective breeding programs; however, error rates in genealogical records have been shown to range from 1-15% within a terrestrial selective breeding program (Bovenhuis and Van Arendonk, 1991, Crawford et al., 1993, Sanders et al., 2006). Genealogical records in aquatic selective breeding programs have the potential to be more erroneous due to large family sizes and difficulties in retaining pedigree throughout the production cycle, particularly in juvenile stages. Genealogical errors lead to inaccurate estimated breeding values (EBVs), a reduction in genetic gain, and inaccurate estimates of inbreeding within the selective breeding program (Banos et al., 2001, Israel and Weller, 2000). The degree to which these inaccuracies affect genetic gains is correlated with the percentage of errors and the length of the selective breeding program. In general, as genealogical errors within the population increase, the number of inaccuracies detected are expected to rise (Banos et al., 2001). In order to reduce these errors, accurate molecular data can be used to correct genealogical records, decrease inbreeding

occurrences, and improve estimates of EBVs and genetic gains (Israel, Weller, 2000; Munoz, et al., 2014; Visscher, et al., 2002).

Considering the potential impact seed supply has on production systems, correctly managing nucleus breeding programs is critical to optimize genetic gain in production systems. With over 4,199,567 metric tons of farmed fish produced in 2014, tilapia are the mostly widely farmed fish globally (FAO, 2017). Tilapia is a fecund freshwater species with a short generation period and sexual maturity reached as young as 6 months of age (Duponchelle and Panfili, 1998). This robust fish is of particular importance in developing countries where it is grown to not only quickly and efficiently meet local protein requirements, but also help improve local job markets (FAO, 2017).

The Abbassa Strain (AS) was initially established by the WorldFish Center (formerly known as ICLARM) in Egypt and relied solely on genealogical data to calculate genetic diversity estimates. The AS originated from a combination of both wild (Aswan, Zawia, and Abbassa) and a hatchery population as founders in 2002, with the objective to increase growth rate using a combination of between and within family selection (Ibrahim et al., 2013, Rezk et al., 2009). This closed selective breeding line produces and maintains approximately 110 full- sibling families per one year generation interval, with matings occurring once per year and each broodstock only contributing to a single mating season. At mating, one adult male and two adult females are held together in a mating hapa. All mating hapas are labeled with a number to identify each family and placed into a single pond. Once one female spawns (i.e. is found to have eggs in her mouth), the male and the other female are removed from the mating hapa, leaving the spawned female to incubate the eggs. Offspring from a single family are then reared in three replicate hapas, each hapa containing 25 – 30 fry, and the remaining fry are kept in the mating

hapa. All hapas are kept closely spaced in one large pond. Once fingerlings are large enough to be tagged with PIT tags, at approximately 8.5 g, the fish are then communally reared until harvest. At harvest, each fish is identified through PIT tags, sexed and weighed. Selection is based solely on the EBV for harvest weight.

The 3.8-7.0% improvement in growth per generation of the AS (Rezk et al., 2009) is much less than the 7.1-15% improvement per generation reported for the GIFT Strain (Eknath, Acosta, 1998, Ponzoni et al., 2011). This brought to question what factors that might explain this relatively slow rate of improvement (for instance, founder quality, pedigree quality, and overall management). For example, in the 4th generation of the AS, new germplasm (2,178 animals) from a sister, less intense, selective breeding line of Nile tilapia were introduced into the AS line. However, only 94 of these individuals were used as broodstock for the next generation. It is unknown how well these introduced individuals integrated into the AS over subsequent generations.

This study used genome-wide molecular information to correct pedigree records and determine the effects on genetic estimates, examine genetic diversity, infer genetic relationships, estimate founder contribution (including, one secondary founder introduction) and inbreeding rates, determine the factors that contributed to the relatively slow rate of improvement observed within the AS, and finally, understand the current state of the strain to assist in determining future strategies for improvement.

2. Methods

2.1 Genealogical analysis

Pedigree records of 28,781 individuals from generation 0 (founders) to generation 11 of the AS were provided by WorldFish. Each broodstock was only used in one breeding period, and there was generational turnover each year. Genealogical analyses were conducted within PEDIG (Boichard, 2002) and ENDOG v4.8 (Gutiérrez and Goyache, 2005) using both the original pedigree and pedigree records corrected by molecular data for generations 10 and 11 (as described in section 2.2). Inbreeding was calculated utilizing the PEDIG program *meuw* (Boichard, 2002). The total number of founders (f) and the effective number of founders (f^e) were calculated via the PEDIG program *prob_orig* (Boichard, 2002). The percentage of each founder genome from both founding events (generation 0 and generation 4) that can be observed in subsequent generations were calculated using the ENDOG v4.8 (Gutiérrez and Goyache, 2005). Analyses were conducted using all recorded individuals as well as only those records of individuals who contributed to the subsequent generation. Differences in results was negligible; therefore, only results from the dataset including all individuals are reported.

2.2 Molecular analysis

2.2.1 Sampling, DNA extraction and genotyping

To the date of these analyses, the AS has been running for 11 generations. Tissue samples were obtained from subset Generations 9-11 as they were the only generations with tissue samples available [122 individuals from generation 9 (G9); 216 individuals from generation 10 (G10); and 54 individuals from generation 11 (G11)]. DNA extractions, genotyping, and co-analysis were conducted by Diversity Arrays Technology (DArT) as described in Lind et al. (2017). To ensure genotype reproducibility, 20% random technical replicates were included internally at DArT.

2.2.2 SNP filtering

Prior to filtering, three individuals with greater than 30% missing data were removed, resulting in a total of 388 individuals (121 G9, 216 G10, and 51 G11) available for subsequent analyses. Single nucleotide polymorphism (SNP) data was then stringently filtered to ensure that only the highest quality and most informative markers were utilized for molecular analyses. A custom Python script, DartQC (<https://github.com/esteinig/dartqc>), was used to select single unique SNPs within clone sequence tags from the dataset, silence genotype calls within each SNP that had less than five read counts, remove SNPs with a minor allele frequency (MAF) less than 0.01, remove SNPs with a call rate less than 0.8, and remove SNPs with an average replication statistic of less than 0.9 (Kjeldsen, et al., 2019; Nayfa, Zenger, 2016).

Raw clone sequences from which SNPs were identified during the DArTseq process were annotated to the available genome assembly for *Oreochromis niloticus* (GenBank Assembly Accession: GCA_00188235.2; Orenil1.1) using a custom Perl script based on NCBI CGI BLAST interface with a 70% minimum sequence identity (Heller-Uszynska et al., 2011). As two linkage groups, LG1 (Palaiokestas et al., 2013) and LG23 (Eshel et al., 2012), are known to have sex-linked loci, SNP markers within either of these two linkage groups, or that were unassigned to a linkage group, were examined for Mendelian Inheritance ratios for sex linked markers, however, no sex linked markers were identified. A total of 6,163 high quality and informative SNPs were retained.

2.2.3 Pedigree analysis

Pedigree records from generation 0 (founders) to generation 11 of the AS were collated from farm records. To test the accuracy of first order relatives in generations 9, 10, and 11 of the AS, comparisons were made between farm pedigree records and the parental assignments based on

166 genotypic data. In order to identify and reduce the potential for miss-assignments in molecular
167 pedigree analysis, a hierarchical pipeline was employed which utilized a random subset of 1,040
168 genome-wide SNPs. Firstly, this hierarchical approach utilized CERVUS v. 3.0, which employs
169 a pairwise likelihood-based parental assignment method to test and identify true parents of
170 individuals from each generation separately. The AS has a yearly mating scheme consisting of a
171 one male to two female ratio within a hapa; however, although unlikely, the close proximity of
172 hapas could results in mating between fish from different hapas via fish potentially jumping or
173 sperm transferred between adjacent hapas. Therefore, respective paternity relationships were
174 tested first using a paternity parentage analysis. Corrected paternal relationships were then
175 utilized in a maternity parentage analysis. A polygamous paternal and monogamous maternal
176 model was utilized with 5% genotyping error rates (Kalinowski et al., 2007). All parental
177 assignments were then confirmed using a second maximum likelihood-based parental assignment
178 program, COLONY v. 2.0.6.3, with a 5% genotyping error rate (Jones and Wang, 2010). This
179 second program was not only used for confirmation, but for its added feature which allows the
180 identification of family groups (i.e. parents and siblings). Only parental matches with greater
181 than 95% probability of correct pair assignments in both Cervus and Colony were retained. Once
182 reassignments were completed, further verification was undertaken using pairwise genetic
183 relationships calculated in GCTA v.1.26.0 (Yang et al., 2011). GCTA v.1.26.0 establishes all
184 individuals analyzed as the base population and defines relatedness so that the average
185 relatedness between “unrelated” individuals is zero (Yang et al., 2011, Powell et al., 2010). As
186 the presence of missing data, low marker MAF and inbreeding distorts the calculation of
187 relatedness, the margin of variation from traditionally accepted relationships (for example, a 50%
188 relatedness between full-siblings, 25% relatedness between half-siblings and a 50% relatedness

between an offspring and its parents) was scaled per individual based on the relatedness calculated to self. Finally, for the family groupings where both parents were genotyped, Mendelian Inheritance errors were also confirmed to be less than 3% using *--mendel* in PLINK 1.9 beta (Purcell and Chang, 2017, Chang et al., 2015). Parental relationships were also tested using the program SEEKPARENTF90 (Aguilar, 2014); however, this program was not as effective if genealogically assigned broodstock had not been genotyped. Depending on the generation, this resulted in 23-79% of the offspring not being tested for parentage. Of those assignments that could be conducted, reassignment rates were within the same range as those obtained in the above described pipeline.

When comparing molecular assignment results to pedigree records, each case was classified into one of four categories: pedigree agreement, reassigned sire or dam, indiscernible sire or dam, or unknown error status (Supplementary Material Table 1). In pedigree agreements, both the molecular and pedigree-based assignments were in agreement. Reassigned sires or dams were those parents who could be correctly assigned using molecular data, but where the molecular and pedigree data did not agree. Indiscernible sires or dams occurred when an offspring's pedigree parents were not assigned to them using molecular data but could not be reassigned since their true parents were not genotyped. Unknown error status was given to those parents whose assignment could not be confirmed due to only partial reassignments of parents that could not be confirmed using the pipeline described above. Assignment mismatches were only identified as 'pedigree errors' (reassigned or indiscernible sires or dams) if both parents could be reassigned, or if both parents had been genotyped and one or more of the assigned pedigree parents was not genetically assigned to the offspring. If both parents were not genotyped, then any partial reassignments (i.e. only a single parent) were not classified as an error since having at least one

unknown parental genotype could introduce ambiguity in relation to the source of the error. Hence, these cases were classified as having an unknown error status. If only one parent was genotyped then any partial reassignments (i.e. only a single parent) were not classified as an error, unless their pairwise genetic relationship confirmed that the parent was not the true parent. To determine whether or not pedigree errors were family based (i.e. genetically assigned parents were siblings to the listed pedigree parent) relatedness was calculated using both molecular and pedigree data for all offspring in generations 10 and 11 who could be assigned at least one parent in generations 9 and 10, respectively, using molecular data. Relationship matrices for pedigree data were built using TASSEL v. 5.0 (Bradbury et al., 2007) for pedigree data and GCTA v.1.26.0 for molecular data (Yang et al., 2011).

2.2.4 Pedigree genetic diversity

Calculations of observed heterozygosity (H_o), expected heterozygosity (H_e), and the number of monomorphic SNPs for generations 9 to 11 were conducted using a Markov Chain (MC) length of 1,000,000 with 100,000 dememorizations in ARLEQUIN v. 3.5 (Excoffier and Lischer, 2010). The average inbreeding coefficient (F) per generation, based on the observed and expected number of homozygous genotypes ($\hat{F}^{ii}$), was established using the command *-ibc* in GCTA v.1.26.0 (Yang et al., 2011). The average multilocus heterozygosity (MLH) for each population was computed using the R package inbreedR (Stoffel, et al., 2016). Finally, the effective population size (N_e) per generation, based on the linkage disequilibrium method (LDN_e), was estimated in NeEstimator v.2.01 (Do et al., 2014).

2.2.5 Pedigree genetic structure

To understand the genetic structure of the AS, individual identity-by-state (IBS) distance matrices from generations 9 to 11 were first constructed using *-distance-matrix* in PLINK 1.9 beta (Purcell and Chang, 2017, Chang et al., 2015). A model-based estimation of ancestry was produced in Admixture v.1.3 which was then used to determine the number of family groups in our subsampling of the AS at a fivefold cross-validation rate (Alexander et al., 2009). Relationships among individuals comprised from the previously constructed distance matrices and admixture proportions were then visualized using NETVIEW v.1.1 at kNN 30 (Neuditschko et al., 2012, Steinig et al., 2016).

3. Results

3.1 Genealogical analysis

While genealogical records were available for generations 1 - 11, tissue samples for molecular analysis were only available for parents (generations 9 and 10) and offspring (generations 10 and 11) of the AS. Subsequently, only those generations with molecular data could be corrected for pedigree errors. On average, 45.5% of AS pedigree records were erroneous per generation (Fig. 1) when comparing parents to offspring (i.e., G9 parents with G10 progeny & G10 parents with G11 progeny). Generation 11 exhibited the highest error rate (54.9%), but with the greatest percentage of its broodstock (Generation 10) genotyped (81.2%) it also had the highest pedigree reassignment rate for identified errors (64.3%), resulting in 35.3% of records in that generation being corrected (Fig. 1). Generation 10, which only had 62.4% of broodstock (Generation 9) genotyped, had a reassignment rate of 52.1% with 25.7% of the errors in that generation corrected (Fig. 1).

A comparison of molecular- and pedigree-based relatedness estimates revealed three clusters that reflect the categories assigned during the investigation of pedigree relationships. As molecular

relationships were sensitive to the proportion of missing SNP data, a range of relatedness between 0.30-0.70 was accepted as a parental match if all other conditions for reassignment were met. This range resulted in less distinct clustering; however, relative groupings were still distinguishable. The cluster with high molecular relatedness (0.32-0.55), but low pedigree relatedness (0.11-0.36) consists of individuals that had parentage reassigned through molecular analysis (Fig. 2). The cluster with high pedigree relatedness (0.57-0.59), but low molecular relatedness (0-0.54) consists of original pedigree assignments that differ from current molecular assignments. The cluster with both high molecular relatedness (0.34-0.69) and high pedigree relatedness (0.57-0.59) consists of those parental relationships that were in agreement between both pedigree and molecular assignments. Of those offspring who had been reassigned, 19.6% of incorrect pedigree assignments were reassigned to siblings of their originally assigned parents and 4.3% of molecularly assigned individuals corresponded to parental siblings based on pedigree data (Fig. 2).

3.2 Founder contributions

Correction of genealogical records was dependent on the available molecular data, and subsequently could only be conducted for generations 10-11. Although genealogical data were shown to have higher error rates than expected, uncorrected genealogical records were used as a benchmark for expected levels of founder contributions, genetic diversity, and inbreeding within the AS. These were then compared to results from molecular data to quantify the effects of pedigree errors.

Despite variations between original and corrected genealogical records, calculations for the total number of founders ($f = 73$) and the effective number of founders ($f^e = 46.6$) were in agreement between both datasets for the most recent generation (G11). Pedigree data indicates that only 83

of the original 201 founders contributed to subsequent generations (Supplementary Material Fig. 1). Of these 83 founders, the family lines of only 53 founders (28 dams and 25 sires) could be traced to corrected generation 11 records with a minimum contribution of 0.1% (Fig. 3). Similar results were also observed in estimates of effective population size (LDNe) for generations 9 (LDNe = 52.6-53.7) and 10 (LDNe = 53.9-54.7). Only the LDNe data of generations 9 and 10 were reported as over 60% of available broodstock were sampled in these generations, whereas a sampling bias occurred in generation 11, with less than 20% of broodstock sampled in this generation. Of the 53 founders, 47 have a traceable contribution of greater than 0.3% to generation 11, which corresponds to the effective number of founders of 46.6 (Fig. 3; Supplementary Material Table 2).

During the 4th generation, 2,178 individuals from another domesticated line were introduced into the AS program, but only 94 individuals were utilized as broodstock for generation 5 of the AS (Supplementary Material Fig. 1). Of these 94 broodstock, only 46 individuals had a traceable genomic contribution of 0.1% or greater to generation 11 of the AS (Fig. 3). Of these 46 secondary founders, all but one individual (1.5%) contributed less than 0.3% to generation 11 genomes. The greatest loss of genetic material from the original 83 founders (14 individuals, 16.9%) occurred in generation 5 after the secondary founders had been incorporated in the AS (Supplementary Material Fig. 1). The number of original founders whose genetic contribution could be traced continued to drop throughout generations 5-9, with a total of 30 (36.1%) founder genomes lost (Supplementary Material Fig. 1). Founder contribution remained consistent throughout generations 9-11. Original founders accounted for 92.0% of AS genetic material (with 34 original founders comprising 84.3% of the AS's genetic composition) and generation 4 secondary founders comprising 7.3% (Fig. 3). Across corrected generations (G9-11), only 0.7%

of the founder contribution remained undetermined due to unassigned pedigrees in corrected genealogical files (Fig. 3).

3.3 Genetic diversity indices

Molecular data reveals that while both H_e and H_o remained relatively constant from generations 9-11 and at a ratio of approximately 1:1 (H_o/H_e ; Table 1). F increased by 0.036 between generations 9 and 10, ($F_{G9}=0.044$; $F_{G10}=0.080$; Table 1). This was followed by a decrease in F in G11 to 0.041; however, this is likely due to sampling bias as only 19.0% of G11 was sampled (Table 1). This was accompanied by an increase in monomorphic SNPs (207) and a decrease in multi-locus heterozygosity between G9 to G11 (3.7%; Table 1).

Based on the original genealogical data, there is a steady increase in inbreeding from founders to generation 11 ($\Delta F=0.003$ -0.019 per generation) with a slight decrease ($\Delta F=-0.010$ in inbreeding in generation 5 due to the new individuals introduced in generation 4; Fig. 4). The average increase in inbreeding per generation in corrected records was $\Delta F=0.004$. Corrected pedigree records exhibited a slight decrease in inbreeding in the corrected generations and generation 4. It should be noted that these corrected records include a higher number of records with one or more missing parents in generations 10 and 11, with PEDIG assuming unknown parents to be new and unrelated individuals (Boichard, 2002). This may result in inbreeding coefficients being underestimated in these generations. Corrected and original inbreeding coefficient estimates based on genealogical records for generation 10 ($F_{original}=0.057$; $F_{corrected}=0.056$) were approximately 25% less than those values estimated using molecular data (G10, $F=0.080$). The variation between original and corrected pedigrees in generation 4 can be attributed to the original exclusion of founder information for those introduced animals (Fig. 4

3.4 Pedigree genetic structure

A total of 27 family lines were identified throughout generations 9-11 using Admixture v. 1.3 and NETVIEW v.1.1 (Supplementary Material Fig. 2). As only 53.3% of available broodstock were genotyped this number is likely underestimated for the total AS genetic stock. Each node, or individual depicted as a circle, illustrates the relatedness of that individual to all other individuals sampled. Colours within the node indicate admixture, with the AS displaying high admixture amongst families (Supplementary Material Fig. 2). The larger the node, the greater the proportion of the population to which that individual is related. Node size of individuals from the oldest available generation (G9) were all approximately equal in size, suggesting that families contributed equally to subsequent generations (Supplementary Material Fig. 2). The thickness of the lines, or edges, connecting individuals represents the genetic distance amongst individuals. In this case, the lines exhibit similar weights, indicating a similar level of genetic distance amongst individuals. Similar edge weights in addition to short distances and high connectivity between individuals suggests that there is a high degree of relatedness within the AS (Supplementary Material Fig. 2).

4. Discussion

This study used genome-wide molecular markers to 1) determine the factors that contributed to the relatively slow rate of improvement observed within the AS, 2) understand the current state of the strain and 3) assist in determining future strategies for improving farm management.

One key issue which has likely attributed to the slow rate of improvement observed within the AS were the high levels of pedigree errors identified within the study. An average total error rate (including both reassigned and indiscernible parents) of 45.5% was observed in pedigree records. At over three times higher than the maximum observed pedigree error rate (1-15%) in other selective breeding programs, this is a significant difference between the AS and other systems

(Bovenhuis and Van Arendonk, 1991, Crawford et al., 1993, Sanders et al., 2006). Discussions with the breeding team revealed a number of issues that occurred over time: such as, multiple naming protocols, unclear IDs, transcribing errors, historical errors (i.e. errors from previous generations causing downstream misidentification) and changes in management. In addition to unforeseen mixing of untagged fish amongst hapas (fine mesh net enclosures) due to the close proximity of hapas and during flooding events. All of these factors could contribute to an accumulated negative effect on the breeding program.

Once detected, 52.1-64.3% of pedigree errors were reassigned to their correct parent pair within this study. These reassignment rates could be improved by genotyping more broodstock from within respective generations so that family groupings with unconfirmed parents can be assigned their true parentage. As pedigree errors remained undetected until molecular data was available, it is suggested that genotyping be incorporated into selective breeding programs as early as possible to not only correct, but to prevent the accumulation of these errors.

With weight estimated to have a heritability of 0.15-0.41 for *O. niloticus* (Khaw et al., 2009), the lower genetic gain observed within the AS of *O. niloticus* (3.8-7.0%; Rezk et al., 2009) compared to the GIFT (Genetically Improved Farmed Tilapia) strain (7.1-15.0%; Eknath and Acosta, 1998, Ponzoni et al., 2011) could also have resulted from a range of factors including differences in environment and the base genetic content of those two populations. However, given the high pedigree error rates, it is quite likely that the management of the AS has attributed to a reduction in selective pressure on the AS. Of these errors, pedigree errors that were family based (19.6%; i.e. offspring assigned to siblings of original pedigree parents) likely contributed less to this reduction in genetic gain than other error types.

Despite these high error rates, the AS's current genetic diversity after 11 generations is still sufficient for the continuation of the breeding program. The AS has on average remained under the 1% increase in inbreeding coefficient per generation deemed acceptable for selective breeding programs (Woolliams, 1994, Bentsen and Olesen, 2002). Additionally, molecular diversity indices do not show a significant drop in observed, expected, or multi-locus heterozygosity amongst generations nine, 10, or 11. The inbreeding coefficient (F) calculated using molecular data for G9 and G11 are similar to the inbreeding coefficients calculated using original and corrected pedigree data; however, molecularly calculated F is 25% greater than pedigree estimates of F . This suggests that pedigree records may provide acceptable estimates for inbreeding rates within the AS; however, these may be an underestimate of the true level of inbreeding due to missing data from individuals with unknown parentage. This is a constraint of PEDIG, which views unknown relationships as new genetic material, the estimates of inbreeding rates are considered to represent the lower bounds of what is observed within the farm stocks (Boichard, 2002).

In order to maintain adequate genetic diversity for genetic improvement and adaptability during dissemination, aquaculture management schemes focus on the initial number of founders in their selective breeding lines and subsequently the number of families retained per generation (Skaarud et al., 2014). However, our analyses have shown that maintaining family lines does not ensure that founder genomes are retained. Of the initial 201 founders only 83 contributed to generation 1, with both pedigree and molecular data indicating that only 53 founder genomes were still present in generation 11. While only an approximation due to pedigree errors, pedigree data revealed that of these 53 founders, only 34 (41.0% of original founders) comprised approximately 84.3% of the AS's genetic composition despite maintaining approximately 110

394 full-sibling families per generation for this line. Considering the majority of the founders for the
395 AS came from wild populations, unequal founder contribution is somewhat expected due to
396 failed spawning, batch drop out, domestication selection, or undesirable trait combinations from
397 untested stock. However, this high loss of genetic material through unequal contributions of
398 founders during generation 1 and generations five to nine of the AS program highlights the
399 importance of preserving unique founder genomes throughout the program in order to obtain the
400 highest level of genetic diversity possible within a closed system under selection. If family
401 numbers rather than founder genomes are monitored, then only those individuals within a family
402 that exhibit the desired traits most strongly will be selected as future broodstock. This
403 encourages genetic gain, but limits genetic diversity within a program as these traits can be
404 traced to a subset of high performing founders. As time progresses, the contribution of these high
405 performing founder genomes continues to increase, as demonstrated in the present study.
406 Consequently, it is suggested that alongside family lines, founder contributions also need to be
407 monitored to avoid rapid loss of genetic diversity when managing selective breeding programs.
408 This should ensure the genetic health and longevity of a selected line in addition to maximizing
409 the adaptation potential of the line in the future.

410 Cross breeding different domesticated lines has been shown to reduce the effects of inbreeding
411 and increase genetic diversity within a selective breeding line (Goyard et al., 2008, Stronen et al.,
412 2017); however, the legacy of these crosses in relation to the maintenance of genetic diversity
413 over time is uncertain. In 2006, 2,178 individuals from another Abbassa mass selection breeding
414 program, whose relation to the AS is unclear, were introduced into the AS, resulting in a 0.004
415 (ΔF) reduction in inbreeding in generation 4 of the AS based on corrected pedigree records.
416 However, this reduction in inbreeding was short-lived since only a few of these individuals were

417 selected as broodstock for subsequent generations and their unique genetic contribution
418 diminished substantially within four years. A number of management considerations could have
419 prevented the loss of these introduced genomes which would have not only reduced inbreeding
420 but may have resulted in further performance improvement after having undergone the same
421 selective pressures as AS animals. For example, the identification of novel founder lines
422 throughout subsequent generations would have allowed selective practices to maintain their
423 germplasm within the line, as well as allowing for selection of high performing individuals. The
424 incorporation of this technique would improve the genetic gain over generations in addition to
425 increasing the retention of germplasm, and therefore minimizing inbreeding and maximizing
426 genetic diversity.

427 Accurate genealogical records are essential in optimizing the genetic gain obtained within a
428 selective breeding program. Any inaccuracies in pedigree information will erode the accuracy of
429 EBVs and diminish the rate of genetic improvement within the line (Banos et al., 2001, Israel
430 and Weller, 2000). If recorded pedigrees are not available, or in question, molecular based
431 parentage analyses provide a practical and efficient means to identify parent and offspring
432 relationships within a selective breeding program. Here once the pedigree of the fish has been
433 ascertained through DNA parentage, marking fish with a physical tag will help maintain identity
434 of the fish from this point.

435 However, there is a trade-off between obtaining false positives and negatives in assignments.
436 Presently, there are three main categories of pedigree reconstruction methods: exclusion
437 methods, relatedness-based methods, and likelihood-based methods (Huisman, 2017). Of these,
438 likelihood-based methods are more powerful in assignment capabilities, although they require
439 more computation time, particularly as the number of markers analyzed increases (Hill, et al.,

2008). Whilst both CERVUS and COLONY utilize a likelihood-based method, there are known limitations to these programs. To reduce computational time, CERVUS only considers pairwise likelihoods to find the most likely parent. However, this can result in close relatives who are not true parent and offspring pairs having a positive log-likelihood ratio, potentially resulting in a false positive (Huisman, 2017). This can be particularly problematic when not all parents have been genotyped. COLONY parental assignments are most accurate when analyses are conducted with highly polymorphic markers; if only less informative markers are available, higher rates of incorrectly assigned parents are observed (Jones, et al., 2010). This can be compensated for by using a higher number of markers, with error rates becoming negligible in COLONY ($<4.5E^{-4}$) when 75 or more SNPs are used for analysis with 40% of parents not genotyped (Huisman, 2017). As such, we found that an integrated workflow including the likelihood-based methods described above, pairwise relatedness estimates and exclusion-based (Mendelian inheritance errors) methods should be utilized to ensure that only reliable parentage relationships are produced.

Conclusions and industry recommendations

The prevalence of pedigree error rates in AS were found to be three times greater than in terrestrial programs (45.5% on average). This high pedigree error rate is likely to have contributed to the low levels of genetic gain (3.8-7.0%) per generation observed within the AS, but did not appear to have a major effect on overall inbreeding levels. While the AS aims to produce 100-120 families per generation, only 28 unique family lines were identified based on genotyping 57.1-63.9% of available broodstock from generations 9-11. An assessment of founder contribution to the AS revealed that only 34 founders comprise over 84.3% of available genetic material within the AS, indicating that founder contribution has been eroded within the

AS. These results suggest a review of the breeding program procedures is necessary, in particular a tightening of the family production process is required to reduce the likelihood of errors; however, the overall genetic composition of the AS population is sound and still provides an acceptable basis for continued breeding. The results also suggest that the inclusion of molecular screening should be introduced, preferably at program inception, to ensure greater pedigree accuracy. It is also recommended that when dealing with closed populations, particularly selective breeding programs, the retention of founder genomes throughout the program should be monitored in addition to family lines.

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Fig. 1 Comparison of genealogical and molecular pedigrees. Genealogically and molecular derived pedigrees were compared and genealogical records were corrected according to molecular records. Records were categorized as 'Pedigree Agreement' (green), 'Unknown Error Status' (i.e. those individuals whose original parents had not been genotyped and who could not be reassigned both parents with at least 95% certainty; yellow), Reassigned Dams (i.e. those offspring who were reassigned a true dam, blue), Indiscernible Dams (i.e. offspring whose genealogically assigned dam was incorrect, but whose correct dam had not been genotyped and therefore could not be assigned, light blue), Reassigned Sires (i.e. those offspring who were reassigned a true sire, black), and Indiscernible Sires (i.e. offspring whose genealogically assigned sire was incorrect, but whose correct sire had not been genotyped and therefore could not be assigned, grey).

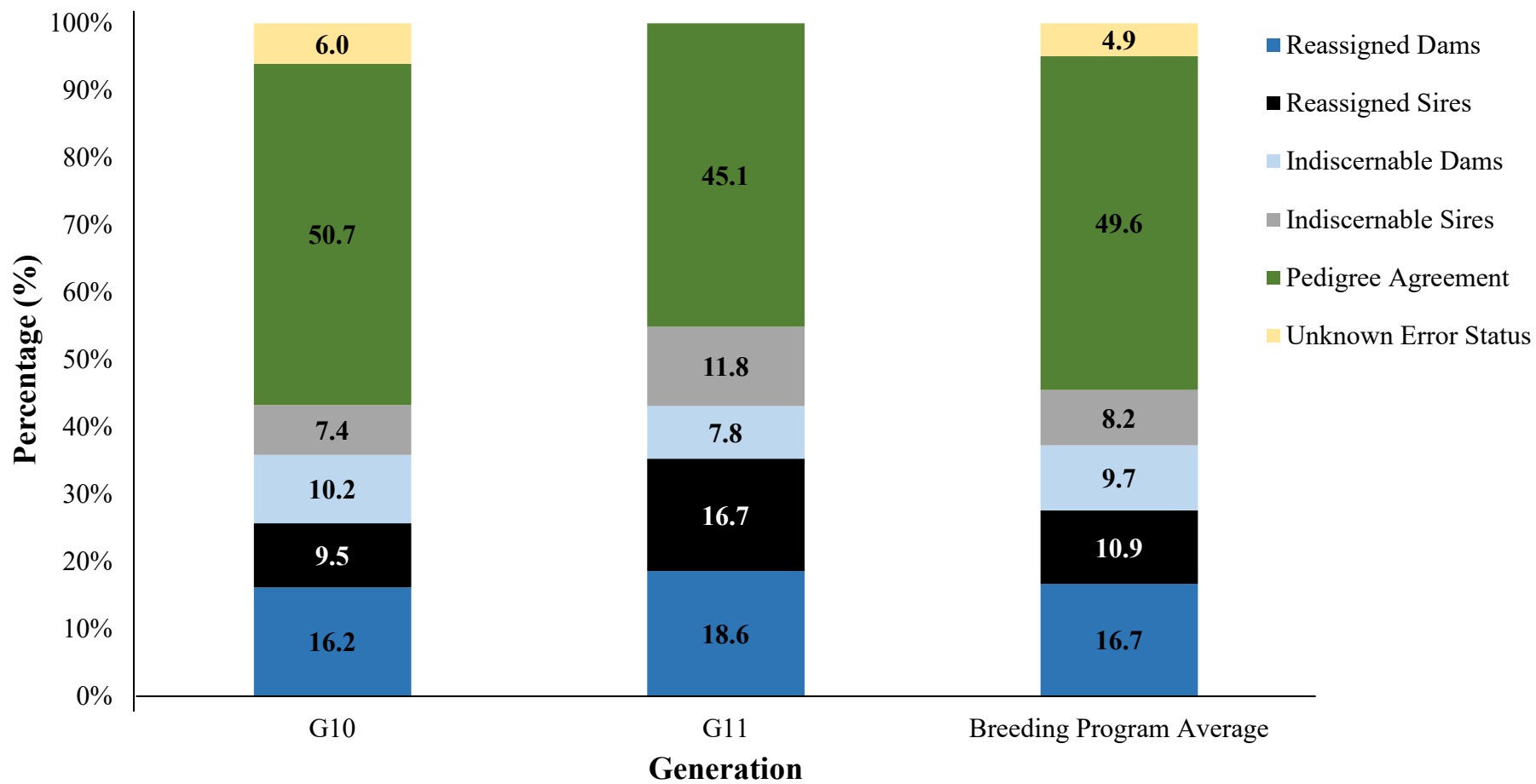


Fig. 2 Comparison of pairwise molecular and pedigree estimated relatedness for parents and offspring Pairings. Colour designations reflect assignment and re-assignment classifications used during pedigree analysis. Comparisons of relatedness are between molecular and pedigree identified parents (whether via molecular assignment, pedigree assignment or agreement between both method) as calculated using molecular data and their original pedigree relationships. Parents whose assignment agreed in both molecular and pedigree estimates are denoted in orange (Agreement). Molecularly assigned parents which did not agree with the pedigree data are denoted in blue (Molecular Assignment). These fell into two distinct groups ranging from 0.1-0.2 in the pedigree relatedness scale (indicating random errors) and 0.3-0.4 in the pedigree relatedness (indicating family-based errors). Those originally assigned pedigree parents not in agreement with molecular data are denoted in grey (Pedigree Assignment). Errors were categorized as random errors from 0-0.1 on the molecular relatedness scale and family-based errors from 0.1-0.5 in the molecular relatedness scale. Please note that molecular ranges vary due to biases resulting from missing data in genetic relationship calculations. Molecular based relationships were scaled per individual based on the relatedness calculated to self. These ranges were then confirmed by the number of Mendelian Inheritance errors detected per molecular family grouping (<3%).

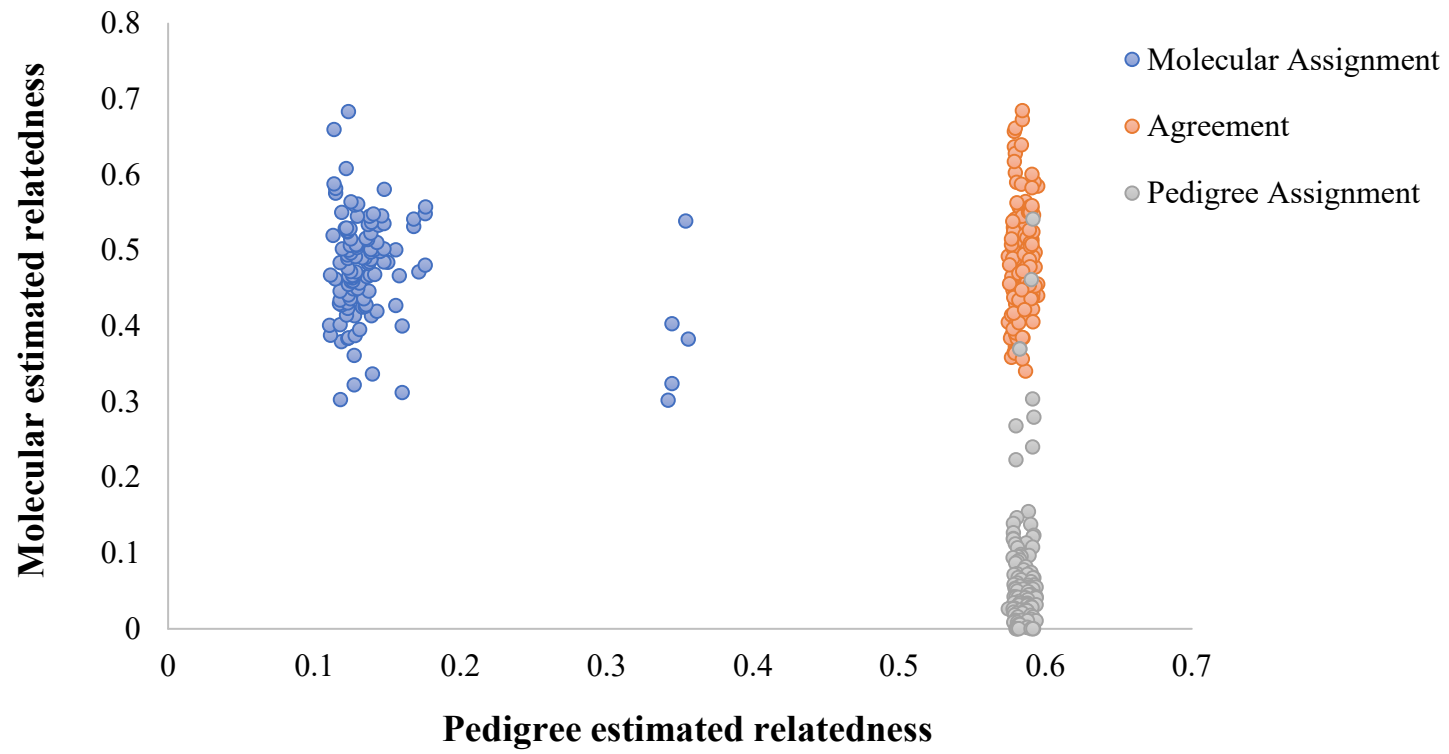


Fig. 3 Genome contributions of founders over 11 generations based on corrected pedigree records. Contributions are based on each founder's average contribution to each generation's genome pool. Contributions have varied over the generations with original founders whose contribution increased in subsequent generations (blue), original founders whose contribution diminished or is negligible over time (tan), contribution of individuals introduced in generation 4 (grey), or unknown contribution due to missing pedigree records (not recorded or due to indiscernible pedigree errors, red). Due to border edge effects, founders with minimal contribution may be indistinguishable from one another (black).

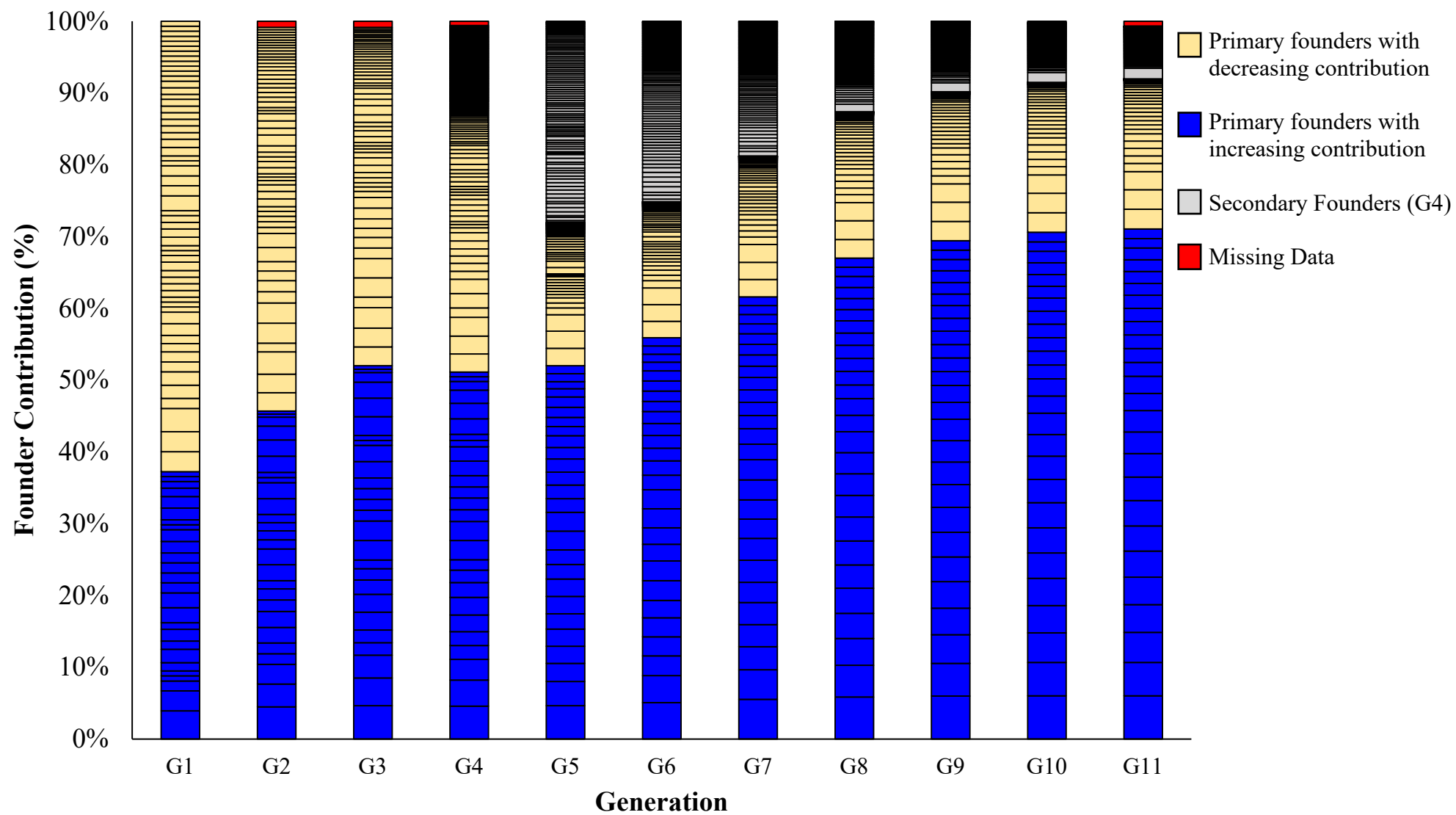


Fig. 4 Inbreeding coefficient (F) for original and corrected genealogical records. Inbreeding based on genealogical records was calculated using the method described by Luo (1992), which utilizes the Cholesky factor of the relationship matrix. Inbreeding coefficients were calculated for both the original pedigree assignments (orange, solid line) and pedigree records corrected using molecular data (blue, dashed line). Error bars represent the standard error of the mean (SE) for each generation. Drops in molecular inbreeding coefficients coincide with the addition of previously undocumented new germplasm (G4) and a bias in the PEDIG program in which individuals with unknown parents are treated as new individuals, underestimating the true level of inbreeding within the program (G10-G11).

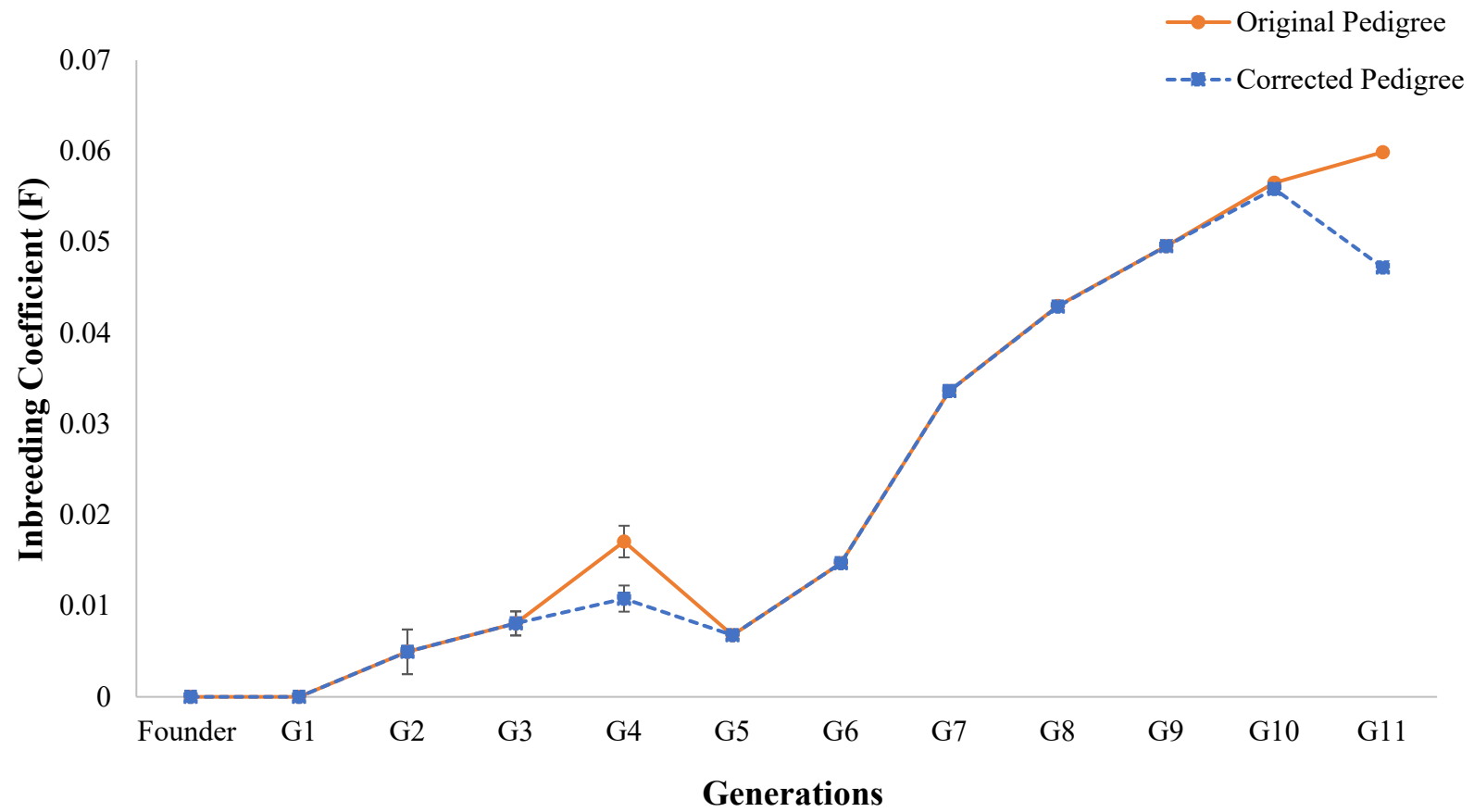


Table 1. Molecular measures of genetic diversity. Expected heterozygosity ($H_e \pm SD$), observed heterozygosity ($H_o \pm SD$), multi-locus heterozygosity ($MLH \pm SD$), inbreeding coefficient ($F \pm SD$), and monomorphic SNPs are presented for generations 9-11 using molecular data.

	G9	G10	G11
H_e	0.225 ± 0.144	0.223± 0.146	0.225 ±0.147
H_o	0.226 ± 0.153	0.221 ± 0.153	0.221 ± 0.157
MLH	0.215 ± 0.025	0.204 ± 0.031	0.207 ± 0.019
<i>F</i>	0.044 ± 0.063	0.080 ± 0.074	0.047 ± 0.020
Monomorphic SNPs	20	15	227